MIL-P-20192D
22 February 1982
SUPERSEDING
MIL-P-20192C
18 November 1975

MILITARY SPECIFICATION

POLE, RANGE, METAL, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of range pole containing two sections of steel tubing. The assembled length of the range pole is 6-1/2 feet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-489	- Enamel, Alkyd Gloss (for Exterior and Interior Surfaces).
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloth.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 6675

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MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-3530	- Thread and Twine; Mildew Resistant or Water Repellent Treated.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-595	- Colors

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The range pole shall consist of two sections of steel tubing, a 3-1/2- to 6-inch hardened steel point, and a carrying case.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3)

3.3 Range pole fabrication. The range pole shall be made of two sections of steel tubing of nominal 1-1/8 inch outside diameter and of nominal 0.032-inch wall thickness. The assembled length shall be 6-1/2 feet, including a hardened steel point permanently fastened to the lower end. The steel point shall be concentric to the centerline of the pole within .050

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inch. The construction of the telescopic sleeve joining the upper and lower sections shall be such that both sections of the pole are the same length. A spring catch, bayonet joint or 5/8 x 11 screw joint shall be provided to lock the two sections rigidly together.

3.4 Carrying case. The two sections of the pole shall be fitted into a two-pocket case made of cotton duck conforming to CCC-C-419, type III, minimum weight per square yard 8.25 ounces for the body of the case and 18.48 ounces for the reinforcing duck at the closed end. The reinforcing duck shall be one piece wrapped around the outside bottom of the case body. The open end shall be provided with a strap, flap, and buckle to completely enclose and hold the sections. The case, with the flap in the closed position, shall be not less than 42 inches long. Each of the two pockets shall have a width and depth of not less than 1.125 inches to accomodate the two range pole sections snugly, but without binding. Webbing conforming to MIL-W-530, type II, class 4, width 3/4 inch, shall be used as binding around all edges, including the pocket openings of the case. The cotton duck shall be dyed light green and treated for mildew resistance in accordance with CCC-D-950, type II, class B, inhibitor (f).

3.4.1 Thread. The cotton thread used to fabricate the carrying case shall be rendered fungus-resistant in accordance with MIL-T-3530, type I, class 1, and dyed light green.

3.5 Identification marking. The pole and the case shall be identified in accordance with MIL-STD-130.

3.6 Treatment and painting. The pole shall be treated and painted in accordance with MIL-T-704, type A, except that the finish coat shall conform to TT-E-489 and the colors shall be as follows: The upper 12 inches of the pole shall be painted red, color No. 11105 or 11136 of FED-STD-595, or orange, color No. 12197 or 12246 of FED-STD-595 and the remainder shall be painted in alternating white and red or orange horizontal bands 12 inches long, except that the point at the lower end shall be painted black. The edges of the rings shall be normal to the axis of the pole.

3.7 Workmanship. All parts, components, and assemblies of the range pole shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformnce inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The range pole and carrying case shall be examined for the following defects:

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101. Dimensions not as specified.
102. Treatment and painting not as specified.
103. Carrying case not as specified.
104. Material not as specified.
105. Lock does not hold two sections rigidly together.
106. Steel point not permanently fastened to the lower end.
107. Steel point not concentric with the pole centerline.

4.5 Inspection of packaging. The preservation, packing and marking shall be examined in accordance with the Quality Assurance Provisions of MIL-S-3534.

5. PACKAGING.

5.1 Preservation, packing and marking. Each pole shall be preserved, packed and marked in accordance with MIL-S-3534. The level of preservation and the level of packing shall be level A, B, or C, as specified (see 6.2)

6. NOTES

6.1 Intended use. The range pole is intended to present a view for an instrument operator sighting a straight, relatively narrow line extending vertically above the point which marks a survey station.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
c. Level of preservation and level of packing required (see 5.1).
d. When the carrying case only is required (see 3.4).

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6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
Army -ME

Review Activity:
DLA - GS

User activity:
Navy - MC

Preparing activity:
Army - ME

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-20192D Pole, Range, Metal, with Case

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ *IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS?* PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. *RECOMMENDED WORDING CHANGE*

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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